

**Forecast accuracy came out
suspiciously high**

**Two LEFT JOINS fixed
every missing month**

ATLIQ HARDWARE GDB0041

MYSQL | Vishnu Ram Sachin | Data Analyst



Problem Statement

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- Yesterday : Fixed running total that broke at the fiscal year boundary
- Today Leadership wants an honest forecast vs actuals comparison per product per month

The Trap ❌

- INNER JOIN on date , customer_code and product_code looks clean but it only keeps month present on both sides

Forecast , Zero sold , Dropped

Sold , Never forecast , Dropped



Requirment

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Keeps every forecasted month ,
even with Zero sales



Keep every sold month ,
even with Zero forecast



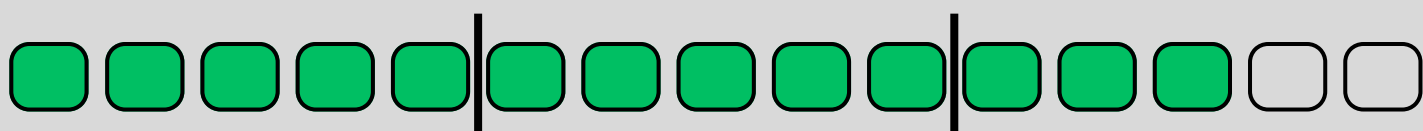
Two LEFT JOINs , one filter
not a single INNER JOIN

The Fix

```
FROM fact_sales_monthly fsm
LEFT JOIN fact_forecast_monthly ffm
  UNION ALL
FROM fact_forecast_monthly ffm
LEFT JOIN fact_sales_monthly fsm
WHERE fsm.product_code IS NULL
```



```
WITH forecast_actual_compare_report AS (  
    SELECT  
        fsm.date,  
        MONTHNAME(fsm.date) AS month_name , fsm.customer_code,  
        fsm.product_code,  
        fsm.fiscal_year,  
        COALESCE(ffm.forecast_quantity, 0) AS forecast_quantity,  
        fsm.sold_quantity  
    FROM fact_sales_monthly AS fsm  
    LEFT JOIN fact_forecast_monthly AS ffm  
    ON fsm.date = ffm.date  
    AND fsm.customer_code = ffm.customer_code  
    AND fsm.product_code = ffm.product_code
```



UNION ALL

SELECT

ffm.date,

MONTHNAME(ffm.date) **AS** month_name, ffm.customer_code,

ffm.fiscal_year,

ffm.product_code, ffm.forecast_quantity, 0 **AS** sold_quantity

FROM fact_forecast_monthly **AS** ffm

LEFT JOIN fact_sales_monthly **AS** fsm

ON ffm.date = fsm.date

AND ffm.customer_code = fsm.customer_code

AND ffm.product_code = fsm.product_code

WHERE fsm.product_code **IS NULL**

)

SELECT

facr.month_name, facr.fiscal_year,

facr.customer_code, facr.product_code,

facr.forecast_quantity, facr.sold_quantity,

(forecast_quantity - sold_quantity) **AS** forecast_error

FROM forecast_actual_compare_report facr ;



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Results

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fact_sales_monthlydim_customerfact_gross_pricefact_forecast_monthlySQL File 11* x

Limit to 50000 rows

14

UNION ALL

15

16

SELECT

17

ffm.date,

18

MONTHNAME(ffm.date) AS month_name, ffm.customer_code,

19

ffm.fiscal_year,

20

ffm.product_code, ffm.forecast_quantity, 0 AS sold_quantity

21

FROM fact_forecast_monthly AS ffm

22

LEFT JOIN fact_sales_monthly AS fsm

23

ON ffm.date = fsm.date

24

AND ffm.customer_code = fsm.customer_code

25

AND ffm.product_code = fsm.product_code

26

WHERE fsm.product_code IS NULL

27

)

28

SELECT

29

facr.month_name, facr.fiscal_year,

30

facr.customer_code, facr.product_code,

31

facr.forecast_quantity, facr.sold_quantity,

32

(forecast_quantity - sold_quantity) AS forecast_error

33

FROM forecast_actual_compare_report facr ;

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Fetch rows:

	month_name	fiscal_year	customer_code	product_code	forecast_quantity	sold_quantity	forecast_error
▶	September	2018	70002017	A0118150101	18	51	-33
	September	2018	70002018	A0118150101	11	77	-66
	September	2018	70003181	A0118150101	9	17	-8

Result 1 xRead Only

Results

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fact_sales_monthlydim_customerfact_gross_pricefact_forecast_monthlySQL File 11* x

Limit to 50000 rows

14 UNION ALL

15

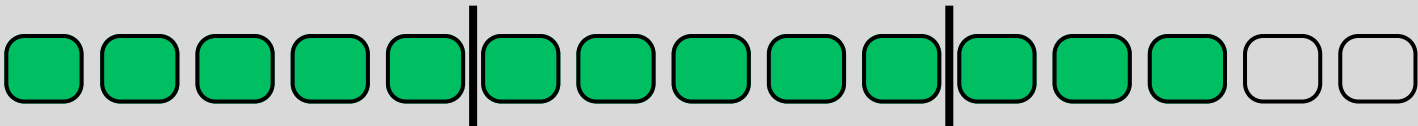
16 SELECT

17 ffm.date,

Result GridFilter Rows:Export:Wrap Cell Content:Fetch rows:

	month_name	fiscal_year	customer_code	product_code	forecast_quantity	sold_quantity	forecast_error
▶	September	2018	70002017	A0118150101	18	51	-33
	September	2018	70002018	A0118150101	11	77	-66
	September	2018	70003181	A0118150101	9	17	-8
	September	2018	70003182	A0118150101	6	6	0
	September	2018	70006157	A0118150101	5	5	0
	September	2018	70006158	A0118150101	6	7	-1
	September	2018	70007198	A0118150101	4	29	-25
	September	2018	70007199	A0118150101	7	34	-27
	September	2018	70008169	A0118150101	7	22	-15
	September	2018	70008170	A0118150101	8	5	3
	September	2018	70011193	A0118150101	5	10	-5
	September	2018	70011194	A0118150101	7	4	3
	September	2018	70012042	A0118150101	0	0	0
	September	2018	70012043	A0118150101	0	0	0
	September	2018	70013125	A0118150101	2	1	1
	September	2018	70013126	A0118150101	2	1	1
	September	2018	70016178	A0118150101	0	1	-1
	September	2018	70022085	A0118150101	12	20	-8

Result 1 xRead Only





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I Vishnu Ram Sachin I Data Analyst

SQL THROUGH ONE DATASET CHALLENGE